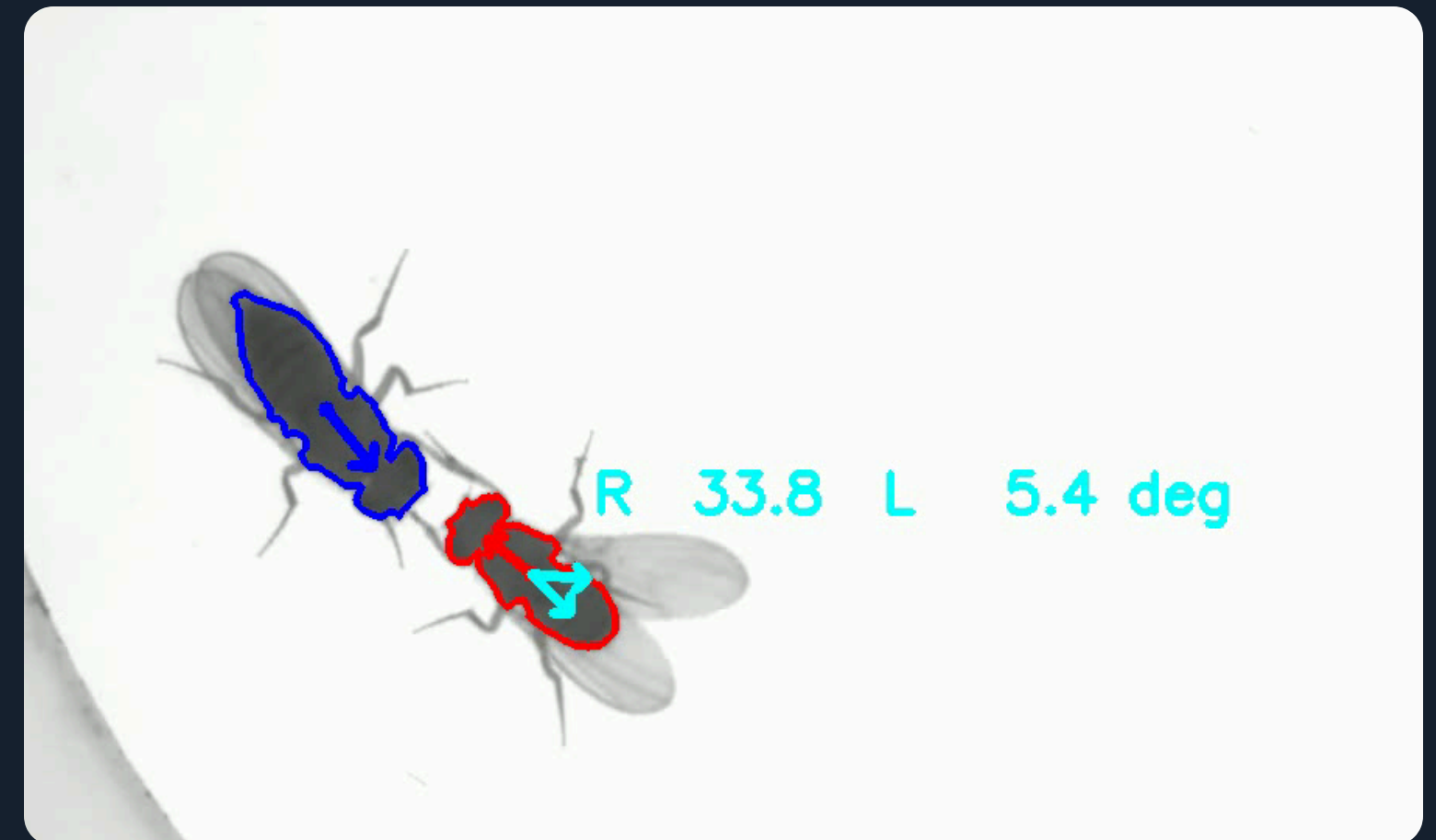


UE24CS352A MACHINE LEARNING · MINI-PROJECT

# Real-time detailed video analysis of fruit flies

A reproduction of Herbst, CS229 Stanford, Fall  
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## PROBLEM STATEMENT

# Track a courting pair of flies faster than the camera records them

### Why real time

Closed-loop experiments react to behaviour as it happens. Long recordings, such as circadian studies, pile up if analysis lags.

### What exists

FlyTracker runs about 10× slower than real time. DeepLabCut and LEAP are fast, but they need a GPU.

### The goal

On 1530×1530 frames, CPU only, report each fly's position, sex and 0–360° heading, and the male's two wing angles.

DATASET

# 326 hand-labelled frames from one 15-minute video

Grayscale courtship video of one male and one female, cropped to the 1530×1530 well. The author labelled frames as points in LabelMe:

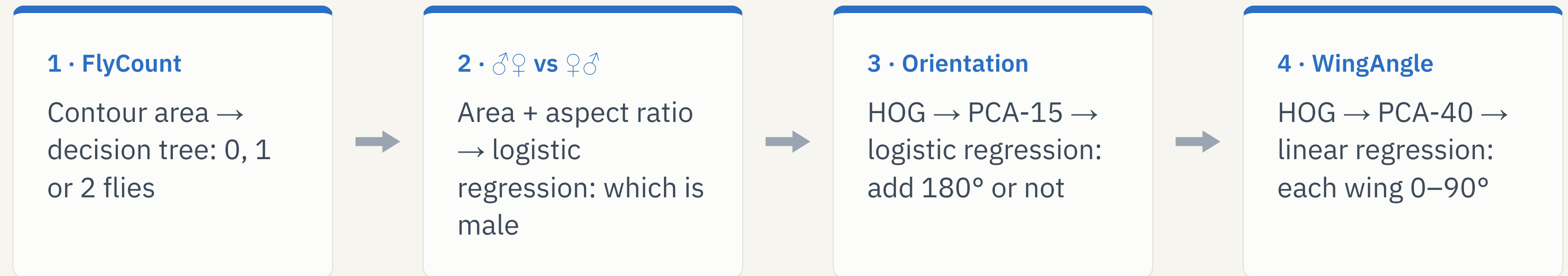
| Points       | Meaning                         |
|--------------|---------------------------------|
| fh · fp · fa | female head, body, abdomen      |
| mh · mp · ma | male head, body, abdomen        |
| mp2 · mw ×2  | wing base and the two wing tips |

| Frame set               | Frames |
|-------------------------|--------|
| Sampled uniformly       | 100    |
| Flies touching          | 33     |
| Running along the wall  | 50     |
| Wing motion, from test4 | 102    |
| Wing motion, from test1 | 41     |

Plus five test clips (9–22 s each) for timing and the demo.

## APPROACH

# Four small models in a chain, no neural networks



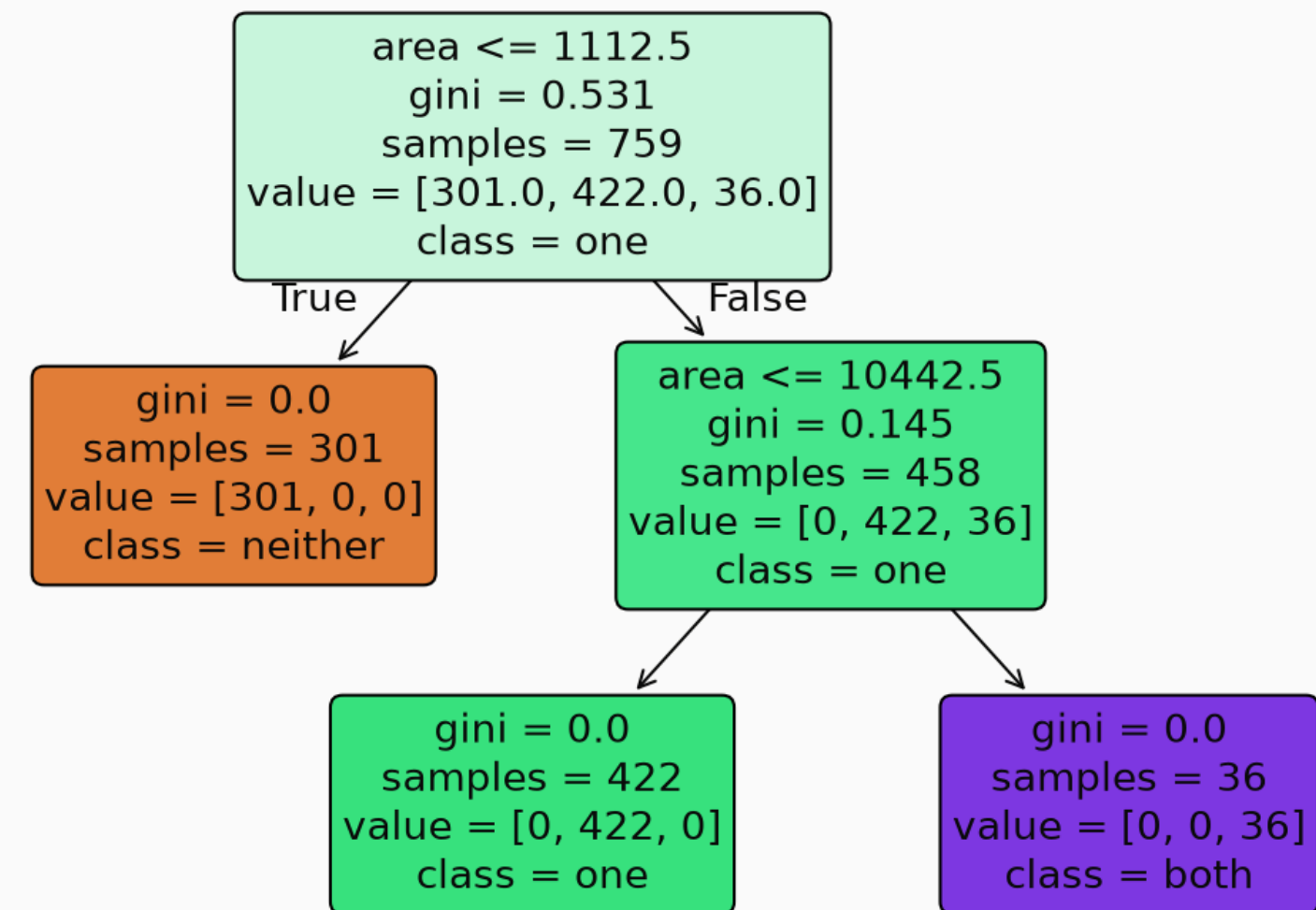
Each stage turns the image into a handful of features and leaves the learning to a simple, fast model. Python with scikit-learn, OpenCV and NumPy, all on the CPU.

# Contour area alone separates noise, one fly and two

- Threshold the frame dark (intensity below 115) inside the well, which leaves only the fly bodies
- One feature per contour: its area
- Decision tree with Gini impurity  $\sum p(1-p)$ , 3 classes

**0.0%** train and test error, as in the paper. The upper split, 10442.5, matches Fig. 4 exactly.

FlyCount decision tree (paper Fig. 4: splits at 1413.5 and 10442.5)



# Females are bigger, so size does most of the work

- The pair is classified jointly: is it ordered male–female or female–male?
- 4 features: area and aspect ratio of both contours. Aspect ratio comes from the ellipse given by image moments.
- StandardScaler, then logistic regression
- Augmentation: swap the pair and flip the label

**0.7%**

test error, same as the paper (1 of 141)

**2.1%**

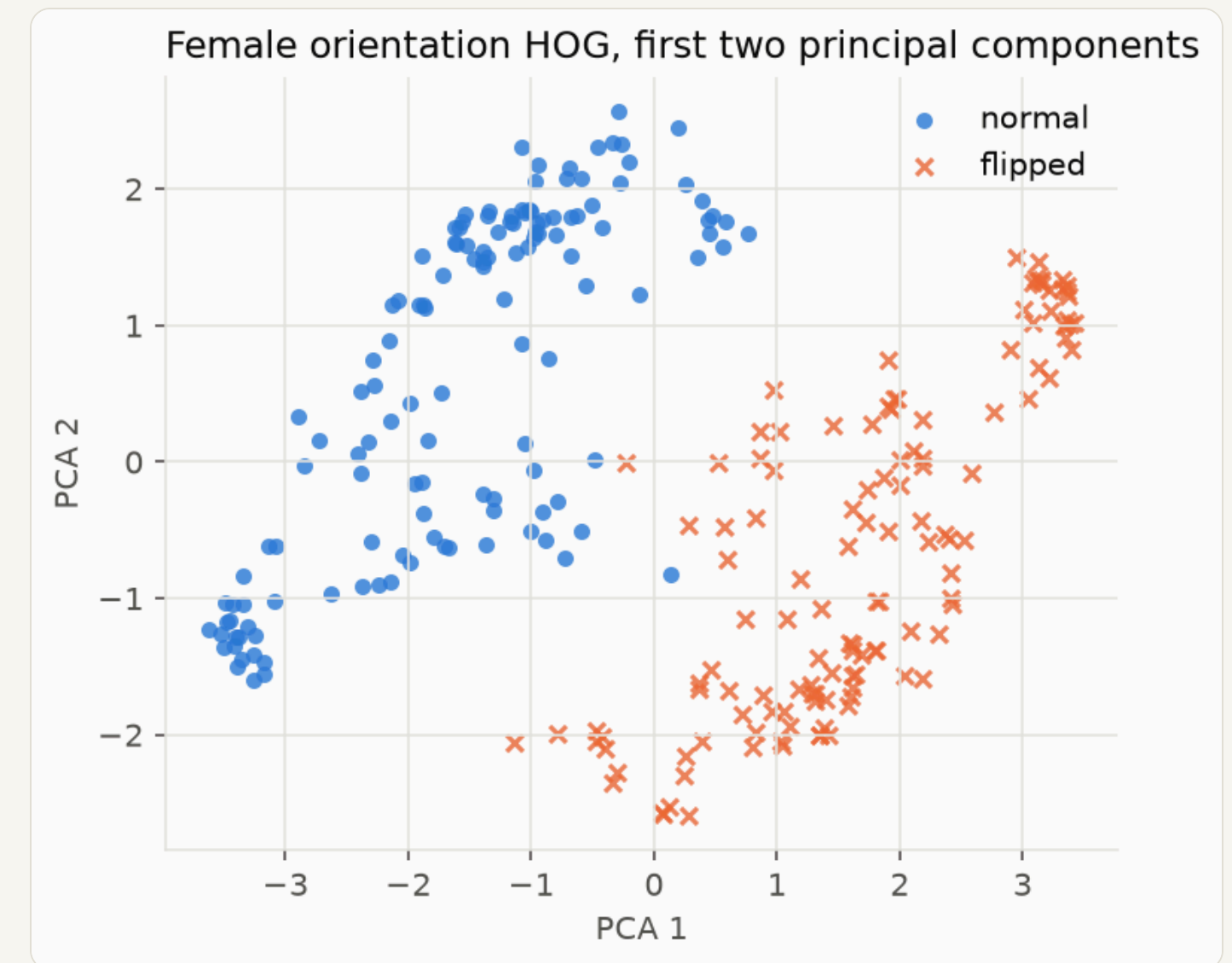
without aspect ratio (poster: 1.4%). Shape helps when flies climb the wall.

### STAGE 3 • ORIENTATION

# Moments give the body axis; a classifier picks the end

- $\theta = \frac{1}{2} \text{atan}(2\mu_{11} / (\mu_{20} - \mu_{02}))$  cannot tell head from tail
- Rotate upright, crop to 128×64, HOG (8×8 cells, 9 bins): a 3780-d vector
- PCA to 15 components, then logistic regression: add 180° or not
- Separate male and female models, augmented by 180° rotation

**0.0%** test error for both sexes, as in the paper

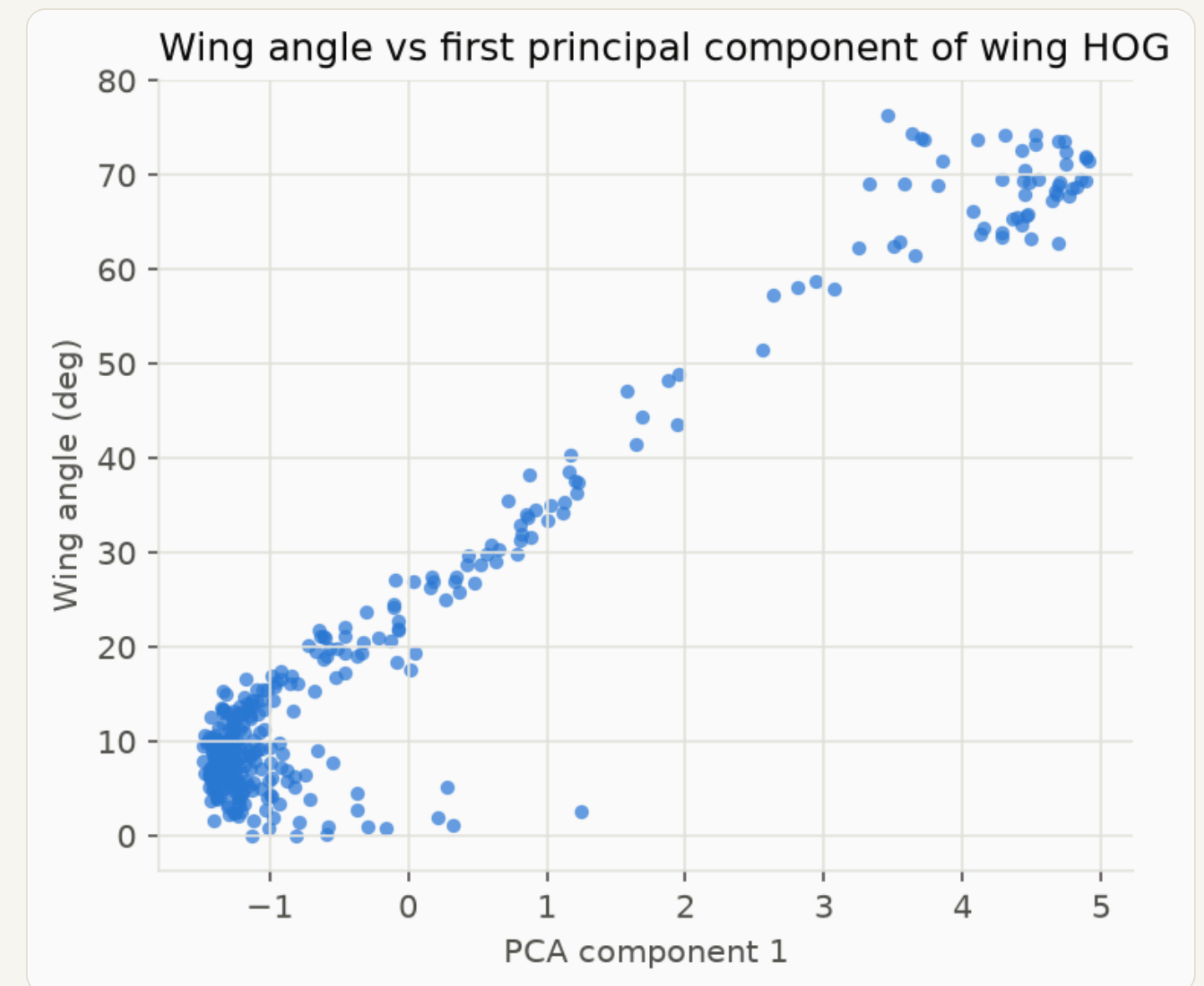


## STAGE 4 · WINGANGLE

# Wing angle is nearly linear in one principal component

- Median blur, threshold, then erosion: this keeps the wings and removes the legs
- Rotate the male upright and crop the right half to 96×80, then HOG: a 3564-d vector
- PCA to 40 components, then closed-form least squares  $\theta = (X^T X)^{-1} X^T y$
- Left wing: mirror the fly and reuse the same model

Test error  $\sigma = 2.18^\circ$  (paper 2.92°). Mean error 0.01°.



IMPLEMENTATION

# One module per stage, one script per result

| File                            | Does                       |
|---------------------------------|----------------------------|
| data.py                         | LabelMe loading, 25% split |
| imgproc.py                      | thresholds, moments, crops |
| flycount.py ...<br>wingangle.py | the four stages            |
| pipeline.py                     | per-frame chain + profiler |
| train.py                        | fit and save all models    |
| evaluate / profile /<br>figures | Table 1, Table 2, figures  |
| demo.py                         | live overlay on video      |

- Only the paper's libraries: scikit-learn, OpenCV, NumPy, Matplotlib, imutils, joblib, tqdm
- Training takes about 40 s and is deterministic (seed 0)
- Values the paper leaves out came from the author's MIT-licensed code. Each one is marked UNSPECIFIED in our code and listed in NOTES.md.
- OpenCV pinned below 5, because 5 dropped HOGDescriptor

# Every model matches or beats the paper's test error

| Model         | Train, ours | Train, paper | Test, ours | Test, paper |
|---------------|-------------|--------------|------------|-------------|
| FlyCount      | 0.0%        | 0.0%         | 0.0%       | 0.0%        |
| ♂♀ vs ♀♂      | 0.0%        | 0.2%         | 0.7%       | 0.7%        |
| ♂ orientation | 0.0%        | 0.0%         | 0.0%       | 0.0%        |
| ♀ orientation | 0.8%        | 0.8%         | 0.0%       | 0.0%        |
| WingAngle     | σ 2.35°     | σ 2.06°      | σ 2.18°    | σ 2.92°     |

Train/test sizes match Table 1 exactly (759/253, 423/141, 358/120, 354/118). The one exception is female orientation: 261 vs 259 training examples.

# Real time on a laptop CPU: 72.5 frames per second

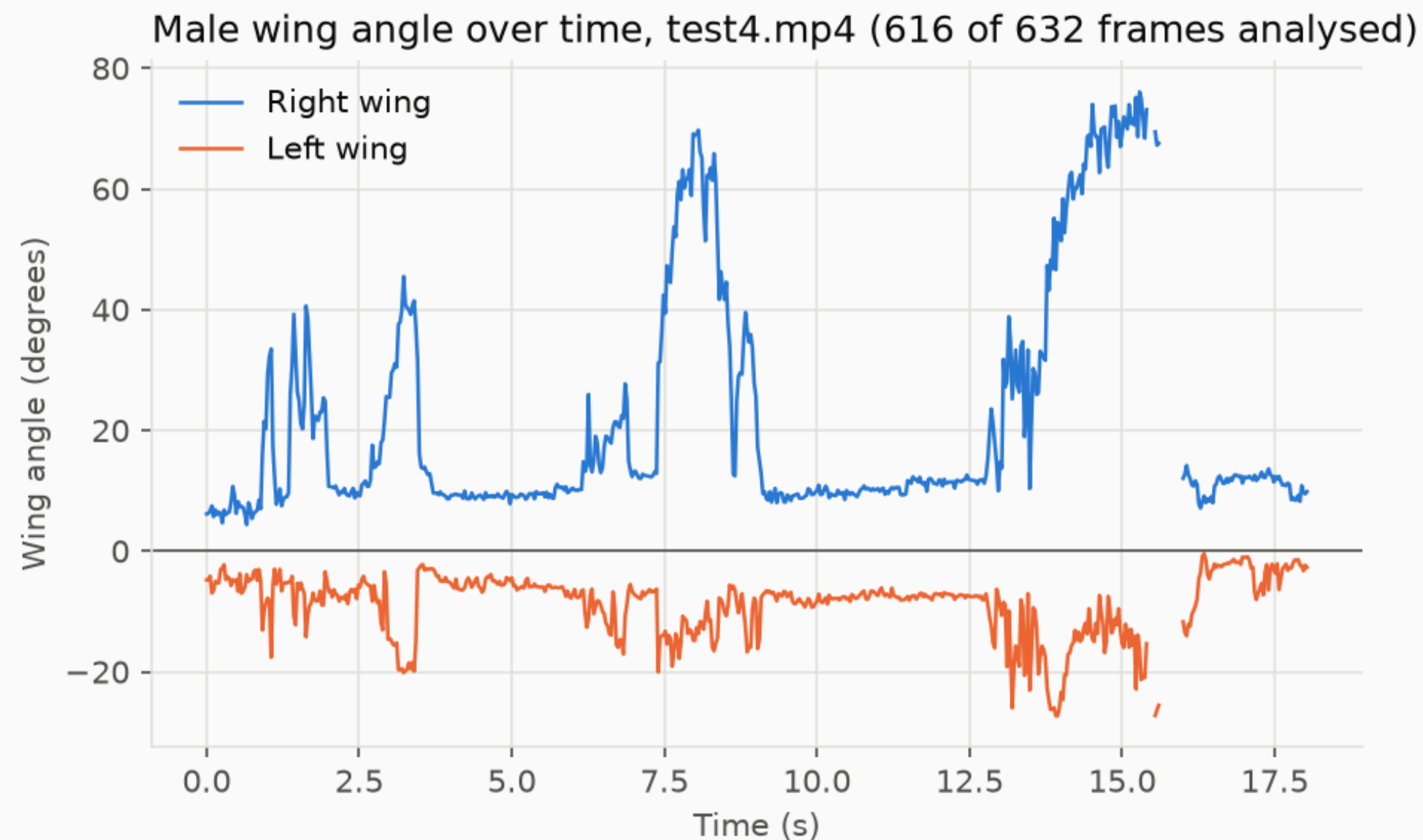
72.5 FPS

ours, 13.8 ms per frame. Paper: 84.0 FPS.

That is 2.4× the 30 FPS source rate. i7-13700HX, 16 GB RAM, no GPU. Median of 5 passes over test4.mp4.

| Step             | Ours   | Paper  |
|------------------|--------|--------|
| Read frame (I/O) | 4.3 ms | 3.9 ms |
| FlyCount         | 4.5 ms | 2.2 ms |
| ♂♀ vs ♀♂         | 0.8 ms | 0.7 ms |
| Orientation      | 1.7 ms | 1.9 ms |
| WingAngle        | 2.5 ms | 3.3 ms |

# Courtship song bursts show up in the wing trace



**−0.69**

right vs left correlation, as plotted.  
Paper: −0.64.

The male mostly extends one wing at a time. 616 of 632 frames were analysed.

# Where the paper and its own numbers disagree

| Topic             | Paper says              | We did, and why                                                                                              |
|-------------------|-------------------------|--------------------------------------------------------------------------------------------------------------|
| Test split        | "one third" held out    | 25%: every Table 1 row is a 25% split                                                                        |
| Leakage           | not discussed           | Split after augmentation, as the author did. A flipped twin can land in test, which flatters the test error. |
| Stage 4 threshold | "adaptive thresholding" | Author's code: well mean – 5                                                                                 |
| Speed-up          | "2.4× faster"           | Poster says 30 FPS, which gives 2.8×. The clips are 35 FPS, which gives 2.4×.                                |

Every gap and choice is logged in NOTES.md in the repo.

## CONCLUSIONS

# Careful preprocessing let simple models run in real time

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### Reproduced

All five Table 1 rows, Figs. 4, 8, 9 and 10, and the real-time claim, on different hardware

### Lesson

Thresholds and crops strip out everything except the variable of interest, so HOG plus 15–40 PCs is enough

### Limits

One video, one random split, augmented twins in the test set, and touching flies are not separated

Live demo: `python src/demo.py -i test4.mp4`